
Report: **AM2024-0074**

BET N₂ sorption measurements on particle samples of natural origin

Short Summary 6

Client:

Curia Germany GmbH
Industriepark Höchst D569
65926 Frankfurt am Main

Contractor:

Fraunhofer-Institut für Silicatforschung ISC
Zentrum für Angewandte Analytik ZAA
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Project Management:

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Documentation Overview

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ISC Offer Number

2210-2024-51000-45-02801-00010 and
2210-2024-51000-45-02801-00019

Order of the Client

signature on offer 2210-2024-51000-45-02801-00010
dated 26 July 2024 and
signature on offer 2210-2024-51000-45-02801-00019
dated 09 September 2024

Place of Performance and Processing Time

Place of Performance

Center for Applied Analytics
Fraunhofer Institute for Silicate Research
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Processing Time

Sample receipt	16.10.2024 & 23.10.2024
Begin of the analysis	25.10.2024
End of the analysis	31.10.2024
Date of report	06.11.2024

Sample Overview

Sample No.	Description
P04	Activated Charcoal Pyrolysate – Batch: MRa329 – Date: 14.10.2024
P05	Activated Charcoal Pyrolysate – Batch: MRa333A – Date: 18.10.2024
P06	Activated Charcoal Pyrolysate – Batch: MRa334A – Date: 18.10.2024

Procedure of Testing

- The powder samples were used as received for the BET N₂ sorption measurements.
- To determine the surface area, the powders were measured in a Quantachrome Autosorb-3B device. The amount of the analysed material was quite little, so the results should be interpreted with a grain of salt.

BET Surface Area

Sample No.	Mass [g]	BET Surface Area [m ² /g]	Langmuir Surface Area [m ² /g]
P04	0.0435	1.76 x 10 ³	2.39 x 10 ³
P05	0.1009	2.09 x 10 ³	5.74 x 10 ³
P06	0.1042	1.86 x 10 ³	2.50 x 10 ³

Final Remarks

- The analyses are carried out by specifically qualified employees and by the use of up-to-date analysis machines. Nevertheless, we are not able to provide guarantee for the correctness of the drawn conclusion.
- The results only refer to the provided and examined samples.
- Sample taking and shipping of the samples to the ZAA took place under the responsibility of the client.
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End of report AM2024-0074 – Short Summary 6

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